Classical Cipher Programs

1. Program to encrypt and decrypt a text file using a **Monoalphabetic Cipher**.
2. Program to retrieve plaintext from a given ciphertext generated using a Monoalphabetic Cipher using **Frequency Analysis Attack**.
3. Program to encrypt and decrypt a text file using the **Playfair Cipher** (5×5 matrix).
4. Program to encrypt and decrypt a text file using the **Playfair Cipher** (16×16 matrix).
5. Program to encrypt and decrypt a text file using the **Vigenère Cipher**.
6. Program to calculate the **Index of Coincidence (IC)** of a given ciphertext file.
7. Program to determine whether a cipher is monoalphabetic or polyalphabetic using **Index of Coincidence**.
8. Program to retrieve plaintext from a ciphertext generated using a Vigenère Cipher using the **Kasiski Test**.
9. Program to estimate the **key length** of a Vigenère Cipher using Kasiski Examination and Index of Coincidence.
10. Program to find the modular inverse $a^{-1} \bmod n$ using the **Extended Euclidean Algorithm**.

OpenSSL Experiments

11. Encrypt and decrypt a text file using **AES in ECB mode** with OpenSSL.
12. Encrypt and decrypt a text file using **AES in CBC mode** with OpenSSL.
13. Compare AES ECB and CBC modes to demonstrate their **security properties**.
14. Evaluate the **avalanche property** of AES encryption:
    - Encrypt plaintext blocks differing by one bit.
    - Compare resulting ciphertexts.
    - Count differing bits and identify changed positions.
    - Repeat the experiment for multiple plaintext pairs.
15. Generate **RSA public and private key pairs** using OpenSSL.
16. **Encrypt and decrypt** a message using RSA.
17. Explain the **parameters** involved in RSA key generation.
18. Generate **message digests** using SHA and MD5 algorithms.
19. Analyze how a **one-bit change** in plaintext affects SHA and MD5 digests.
20. Generate a **self-signed digital certificate** using OpenSSL.
21. Explain the **role and importance** of digital certificates in secure communication.

NMAP Experiments

22. Use Nmap to identify **open, closed, and filtered ports** on a host.
23. Use Nmap to detect **services and software versions** running on target hosts.
24. Use Nmap for **operating system detection**.
25. **Save Nmap scan results** into a file.
26. Explain the **purpose and applications** of the Nmap tool.

Wireshark Experiments

27. Explain the **purpose and applications** of the Wireshark tool.

28. **Capture TCP packets** and save them into a packet capture file.

29. Find the **sequence number** of a TCP SYN packet and the corresponding SYN+ACK packet.

30. Find the **time interval** between two consecutive HTTP GET requests.

31. List and analyze all **headers** of a TCP packet.

32. Identify at least **four protocols** from captured network traffic.

Frequency Analysis & Statistical Cryptanalysis

33. Generate the **frequency distribution** of characters in a ciphertext.

34. Use frequency analysis and **Index of Coincidence** to classify ciphertexts.

35. Perform **ciphertext-only attacks** on substitution ciphers using statistical analysis.

IPTABLES Experiments

36. Configure IPTables rules to **block incoming traffic** from a specific IP address.

37. Configure IPTables to **allow only SSH and HTTP** services while blocking all other incoming traffic.

38. Use IPTables to **restrict access** to a specific port.

39. **Save and restore** IPTables firewall rules.

40. Monitor and list **active IPTables firewall rules** with packet statistics.